

SHUBHAM RODE

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PROFESSIONAL SUMMARY

E&TC graduate driven by precision and real-time performance, specializing in automotive embedded firmware, CAN bus diagnostics, and microcontroller integration to engineer reliable vehicle communication systems.

EDUCATIONS

K.K. Wagh Institute of Engineering Education and Research | Nashik Nov 2022 - Jun 2026

B.Tech in Electronics & Telecommunication

- **Relevant Coursework:** Microcontrollers & Microprocessors, Real-Time Operating Systems (RTOS), VLSI, Embedded Systems, Internet of Things (IoT)

SKILLS

Languages :	C, Embedded C, Python, C++
Microcontollers & Boards :	ATmega128, STM32, ESP32, Raspberry Pi, PIC18F4550, Arduino, Mega 2560
Embedded Tools & IDEs :	STM32CubeIDE, MPLAB X, Arduino IDE, Proteus, MATLAB, Keil µVision, Atmel Studio, Simulink, SavvyCan
Sensors & Actuators :	BMP180, MPX4250, RFID, Ultrasonic, LDR, DHT11, Servo Motors, Stepper Motors (NEMA 17)
Communication Protocols :	UART, I ² C, SPI, CAN, RF, Bluetooth, Wi-Fi
Soft Skills :	Problem Solving, Project-based Learning, Team Collaboration, Adaptability

WORK EXPERIENCES

Renewable Energy & EV Operations Coordinator | Internship Jan 2026 - May 2026
Aviro Energy India Pvt Ltd On-site • Pune

Led system-level layout and infrastructure design for EV charging stations and solar power grids. Conducted technical site analysis to identify optimal hardware placement, power routing, and load distribution for high-voltage setups. Coordinated the integration of EVSE (Electric Vehicle Supply Equipment) components and solar inverters, managing diagnostic data and project workflows through the I-SMART Portal and Bitrix24 CRM to enhance vehicle-to-grid power networks.

Technologies / Skills Used : Infrastructure Layout Design, EVSE Integration, Power Routing, High-Voltage Safety, I-SMART Portal, Bitrix24 CRM

Embedded System Engineer Intern | Internship Dec 2024 - Jan 2025
Samarth Solutions Remote • NIT Bhopal

Led the comprehensive electromechanical design and assembly of a custom 3D printer, integrating an Arduino Mega 2560 and RAMPS 1.4 shield to achieve high dimensional print precision of ± 0.1 mm. Conducted rigorous hardware calibration of NEMA17 stepper motors and A4988 drivers to optimize multi-axis movement and mechanical load distribution. Coordinated hardware integration with Marlin firmware, tuning heat bed PID settings to reduce thermal fluctuations by 15% and ensure system stability for continuous fabrication cycles of up to 11 hours.

Technologies / Skills Used : Arduino Mega 2560, Marlin Firmware, RAMPS 1.4, NEMA17, MK8 Extruder, PLA Filament

PROJECTS

CAN-Sense: ECU Diagnostic and Traffic Monitor Apr 2026 - May 2026
System designer

Engineered a real-time CAN bus diagnostic monitor utilizing MCP2515 controllers to intercept ECU traffic at 500 kbps. Successfully decoded up to 1,000 frames per second with 99.9% logging accuracy.

Technologies / Tools Used : MCP2515 CAN Controller, STM32/ATmega128, SPI Protocol, CAN Bus (ISO 11898), Embedded C, ESP32, Arduino Uno, RapberryPi 5

NetBridge: Lightweight Layer 3 NAT Router Feb 2026 - Mar 2026
Firmware Developer

Engineered an ESP32-based Layer 3 NAT router to enhance Wi-Fi coverage by 35%. Achieved a consistent throughput of 5 Mbps for 5 concurrent devices and reduced packet loss by 25% through lwIP stack optimization.

Technologies / Tools Used : ESP32, lwIP Stack, TCP/IP, Network Address Translation (NAT), DHCP, Wi-Fi 802.11 b/g/n

BharatVote - Smart Facial Recognition-Based Voting System Jul 2025 - Oct 2025
Team leader

Engineered a Raspberry Pi smart voting system using Python and OpenCV, achieving 98.5% authentication accuracy and verifying voters in under 1.2 seconds by reducing processing latency by 40%.

Technologies / Tools Used : Raspberry Pi 5, OpenCV, Face recognition, Thonny, Python 3.11, I2C protocol

TireGuard - Integrated Nitrogen Tank System Sep 2024 - Oct 2024
Team Leader

Engineered an automated nitrogen gas inflation system utilizing an ATmega128 and MPX4250 sensors. Triggered solenoid valves in under 50 milliseconds to maintain tire pressure with ± 0.5 PSI precision.

Technologies / Tools Used : ATmega128, Embedded C, MPX4250 Pressure Sensor, 10-bit ADC, Solenoid Valve, Nitrogen Gas, Atmel Studio, NRF24L01